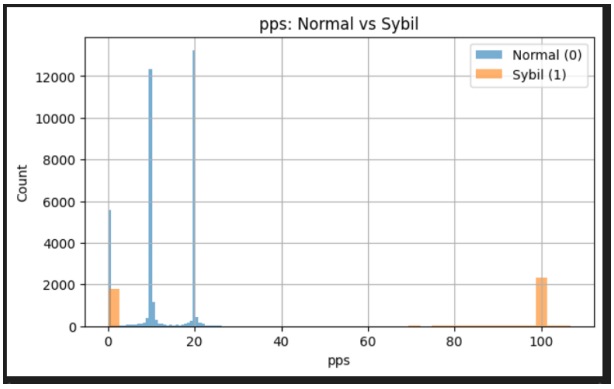
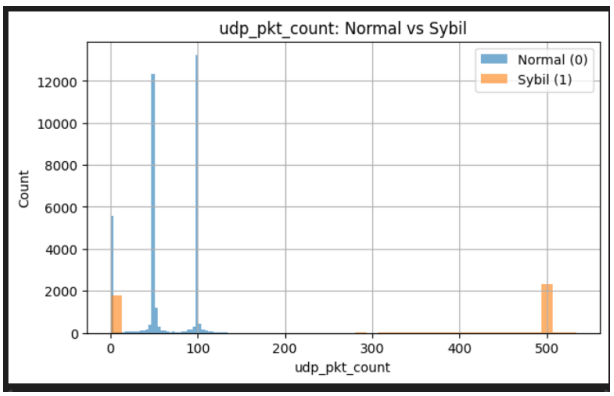
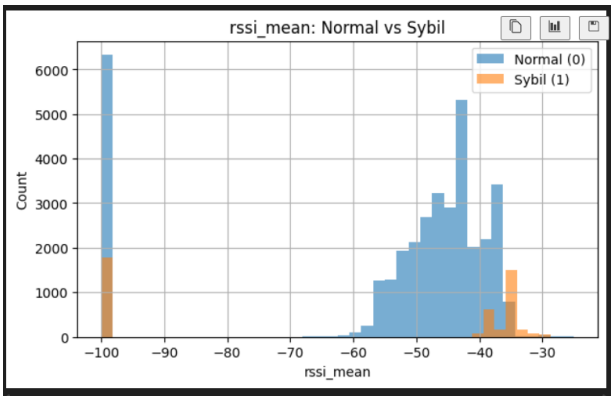
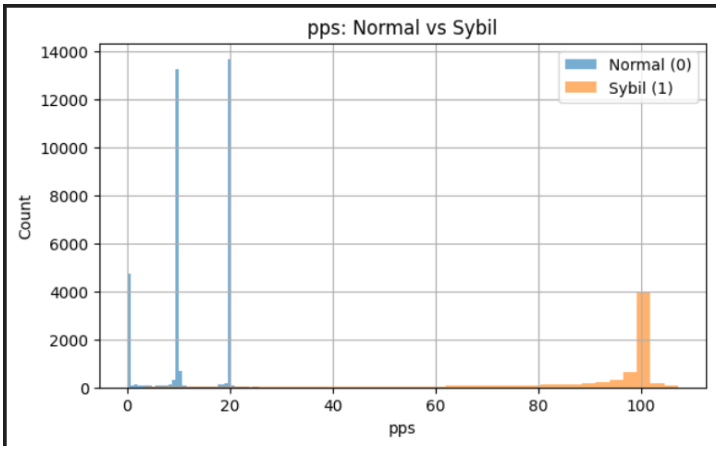
HIGH SIGNAL, HIGH RATE

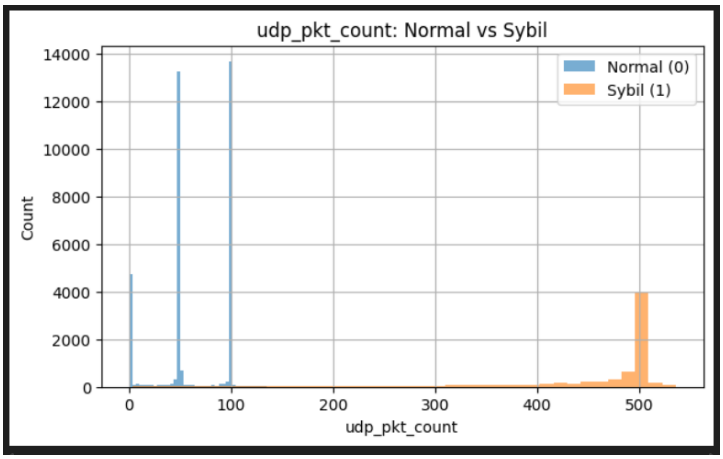


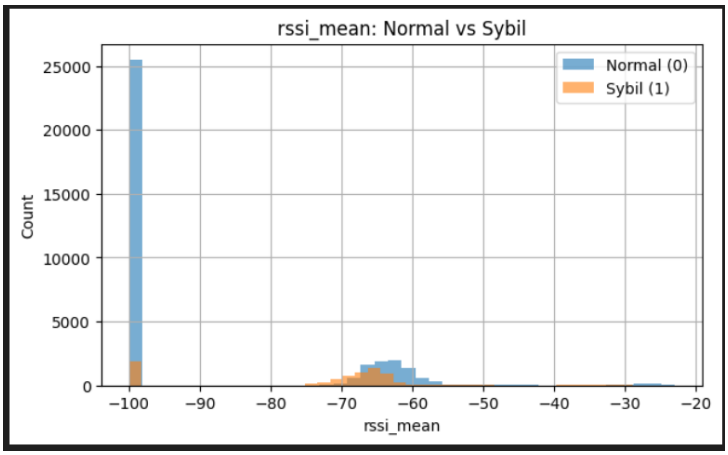




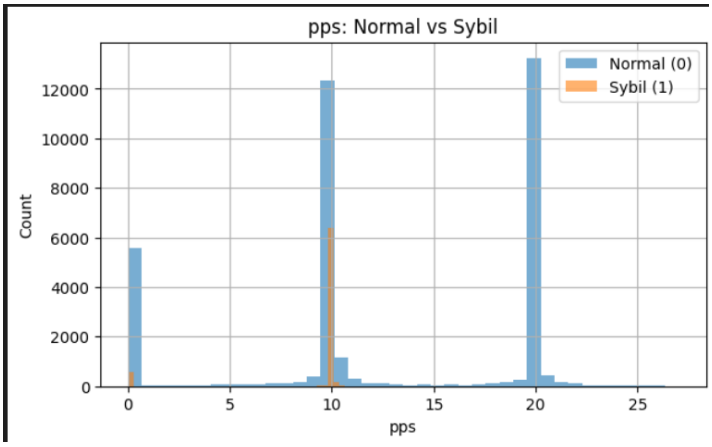
HIGH RATE, LOW SIGNAL

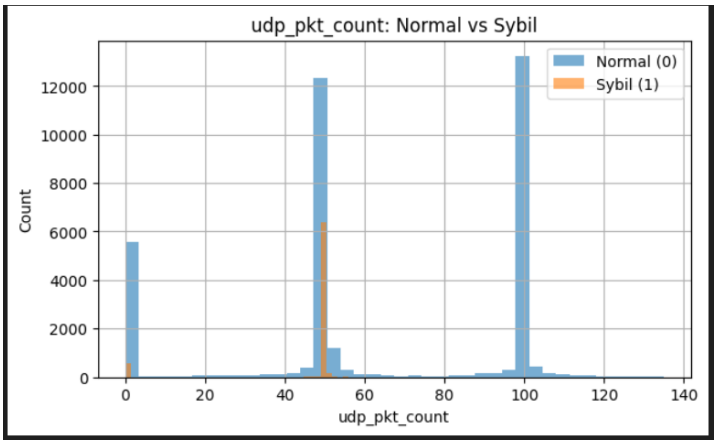


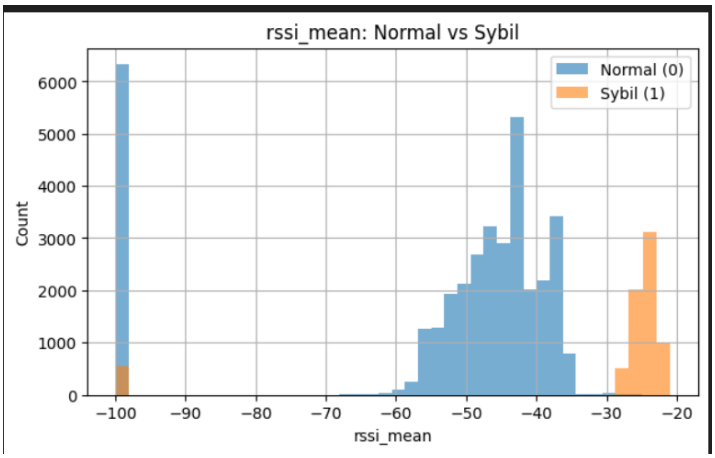




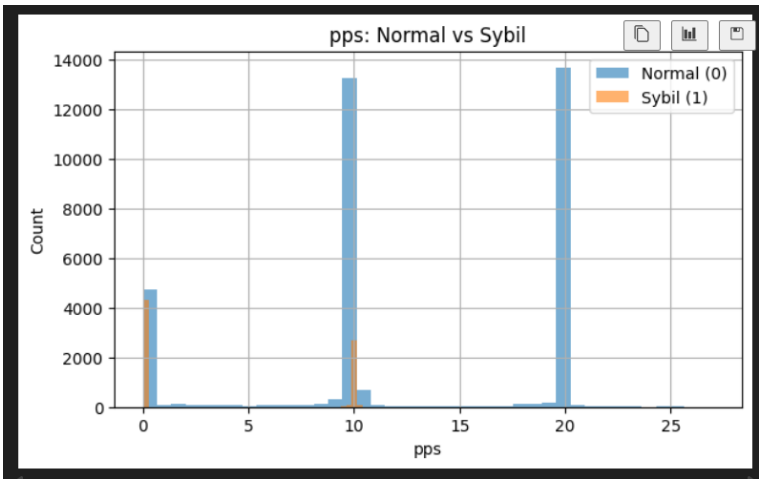
LOW RATE, HIGH SIGNAL

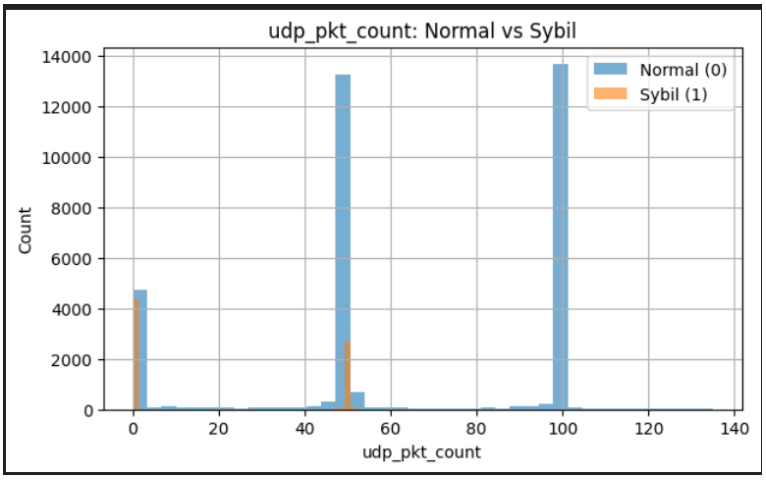


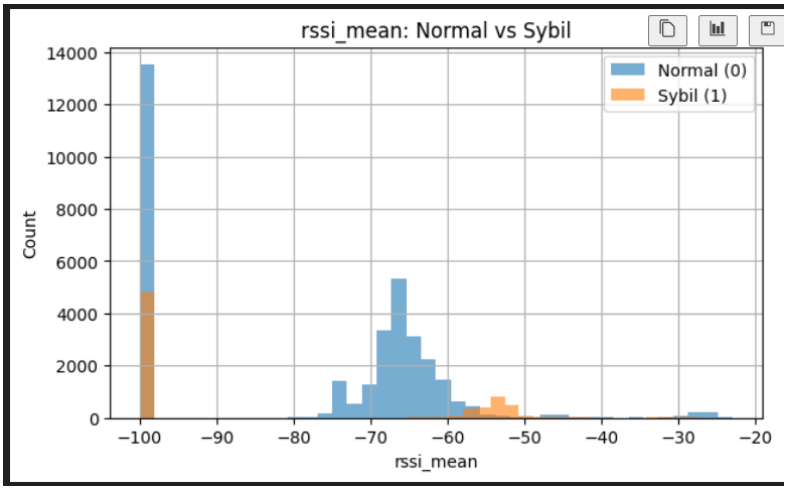




LOW RATE, LOW SIGNAL

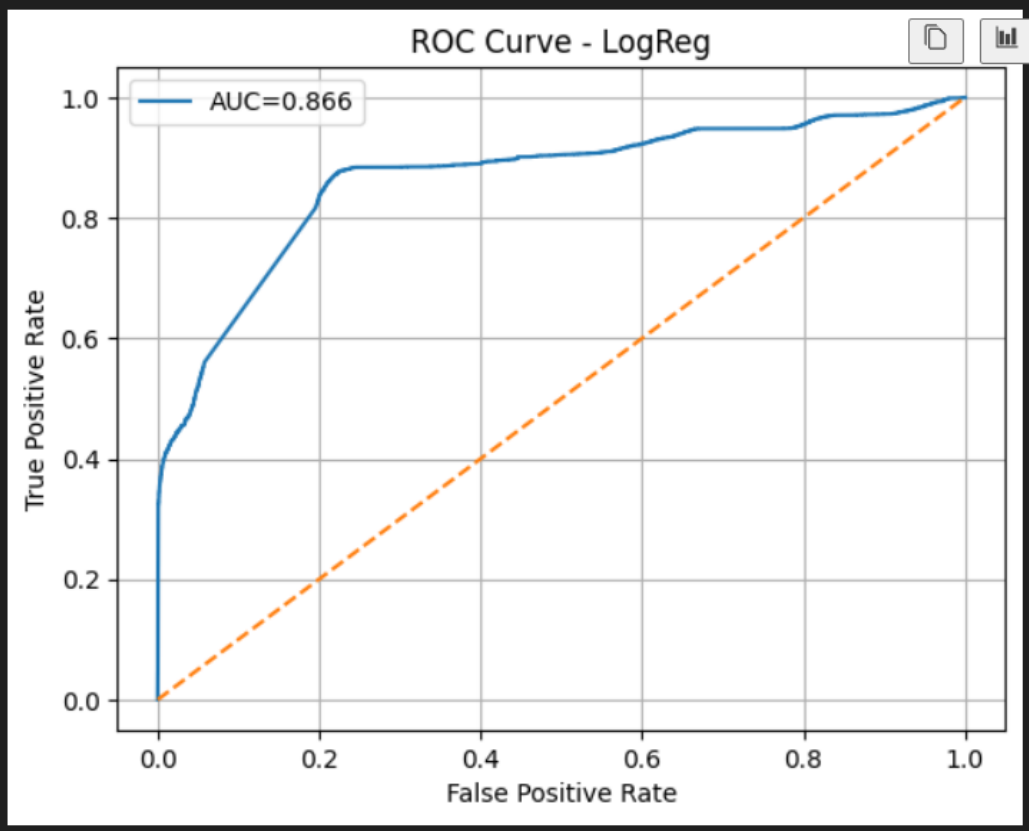

pps: Normal vs Sybil
Normal (0)
Sybil (1)
Count
pps
0
2000
4000
6000
8000
10000
12000
14000
0
5
10
15
20
25

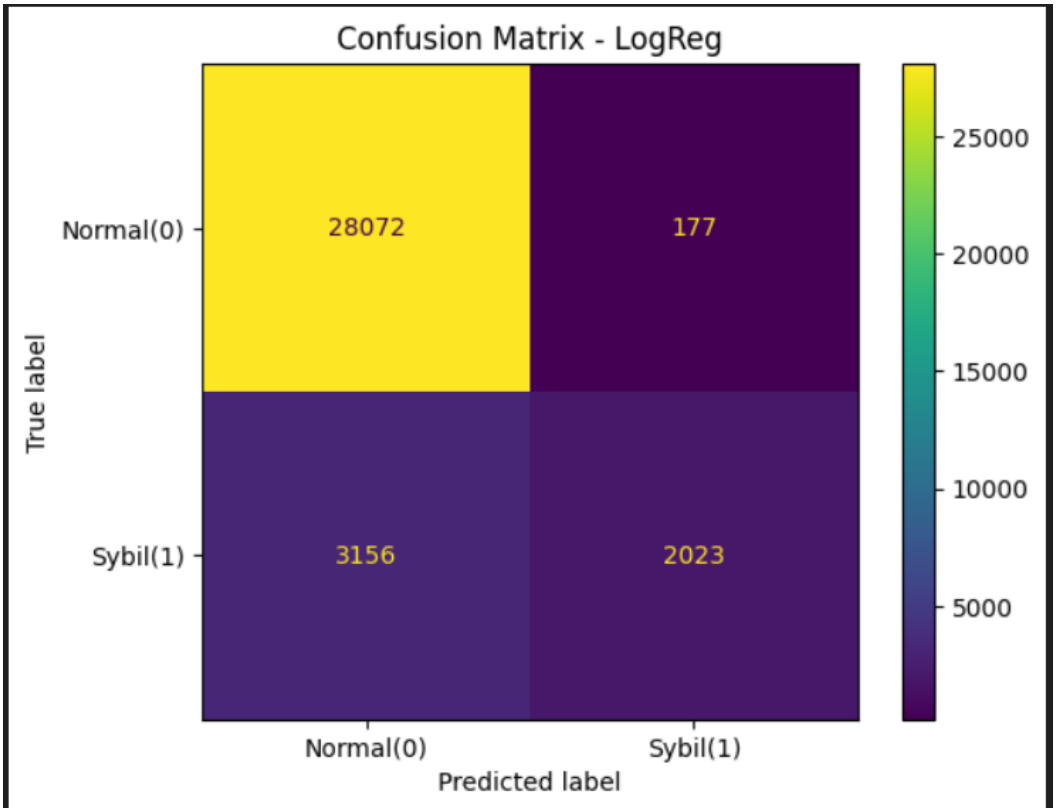

udp_pkt_count: Normal vs Sybil
Normal (0)
Sybil (1)
Count
udp_pkt_count
0
2000
4000
6000
8000
10000
12000
14000
0
20
40
60
80
100
120
140


rssi_mean: Normal vs Sybil
Normal (0)
Sybil (1)
Count
rssi_mean
0
2000
4000
6000
8000
10000
12000
14000
−100
−90
−80
−70
−60
−50
−40
−30
−20

LogReg

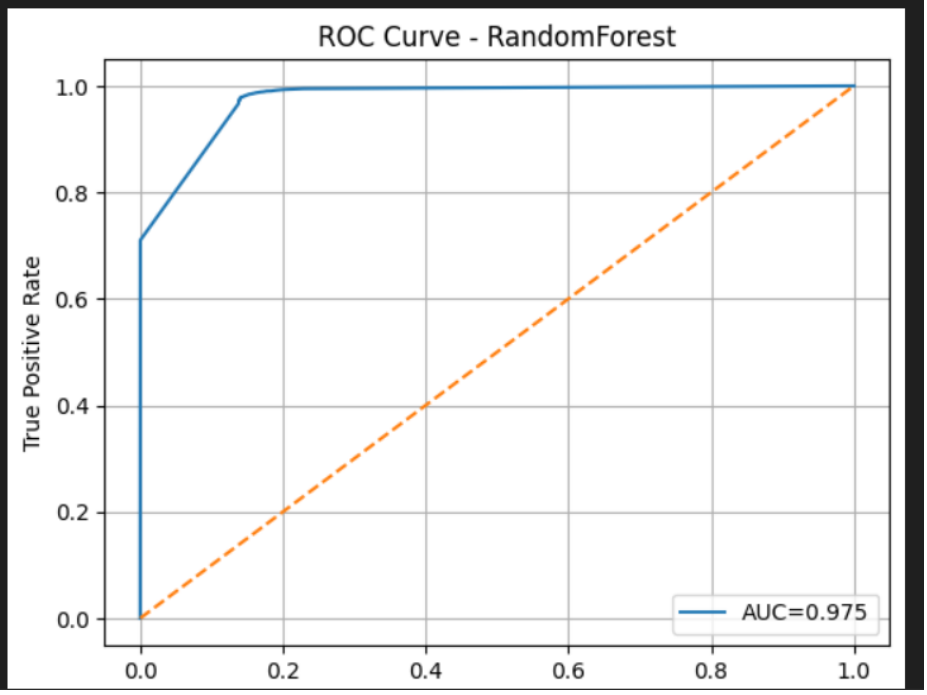
```
MODEL: LogReg
              precision    recall  f1-score   support

           0     0.8989    0.9937    0.9440     28249
           1     0.9195    0.3906    0.5483      5179

    accuracy                         0.9003     33428
   macro avg     0.9092    0.6922    0.7461     33428
weighted avg     0.9021    0.9003    0.8827     33428
```

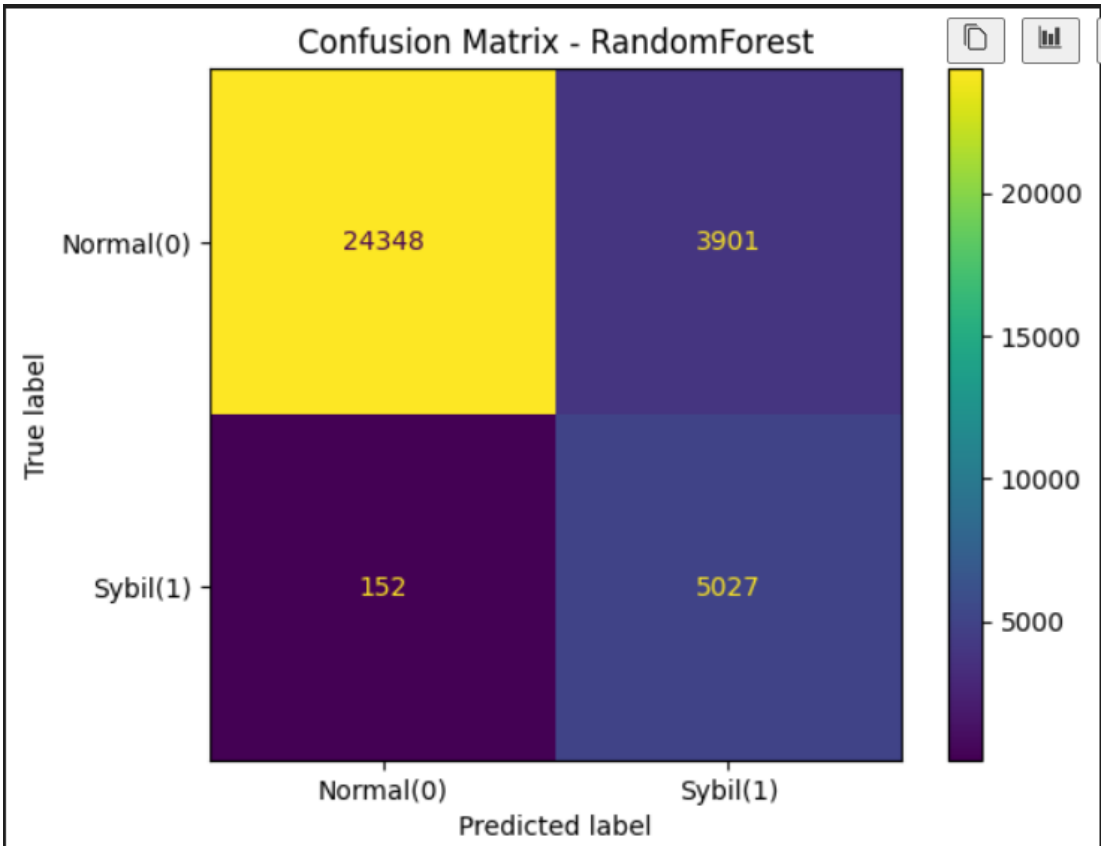




MODEL: RandomForest

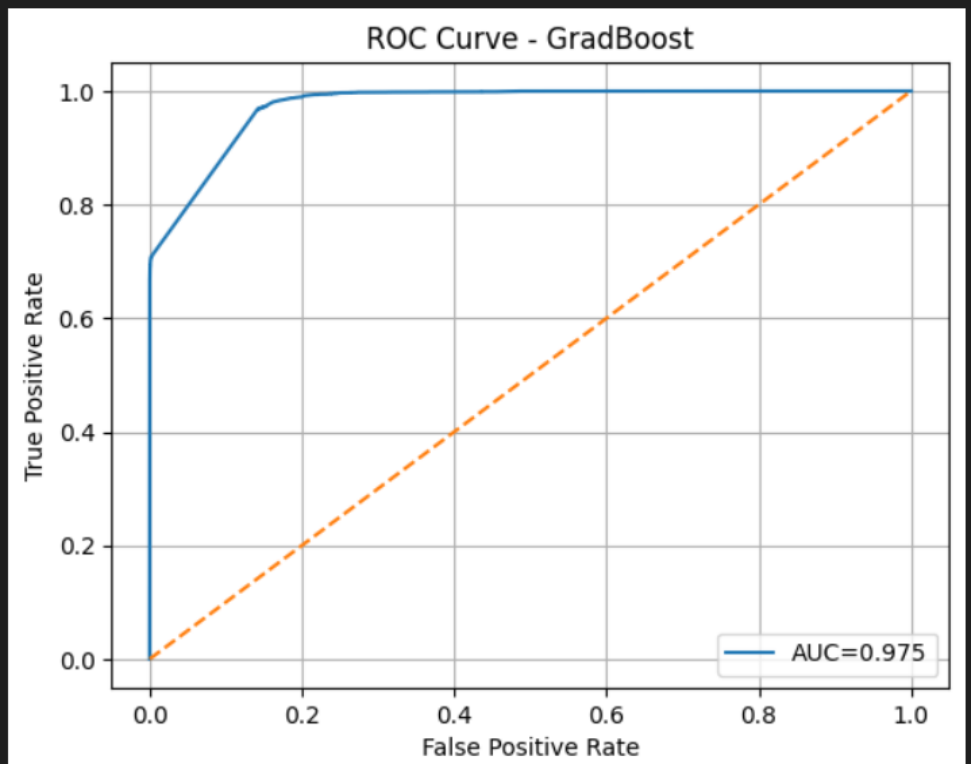
```
==============================
MODEL: RandomForest
              precision    recall  f1-score   support

           0     0.9938    0.8619    0.9232     28249
           1     0.5631    0.9707    0.7127      5179

    accuracy                         0.8788     33428
   macro avg     0.7784    0.9163    0.8179     33428
weighted avg     0.9271    0.8788    0.8906     33428
```

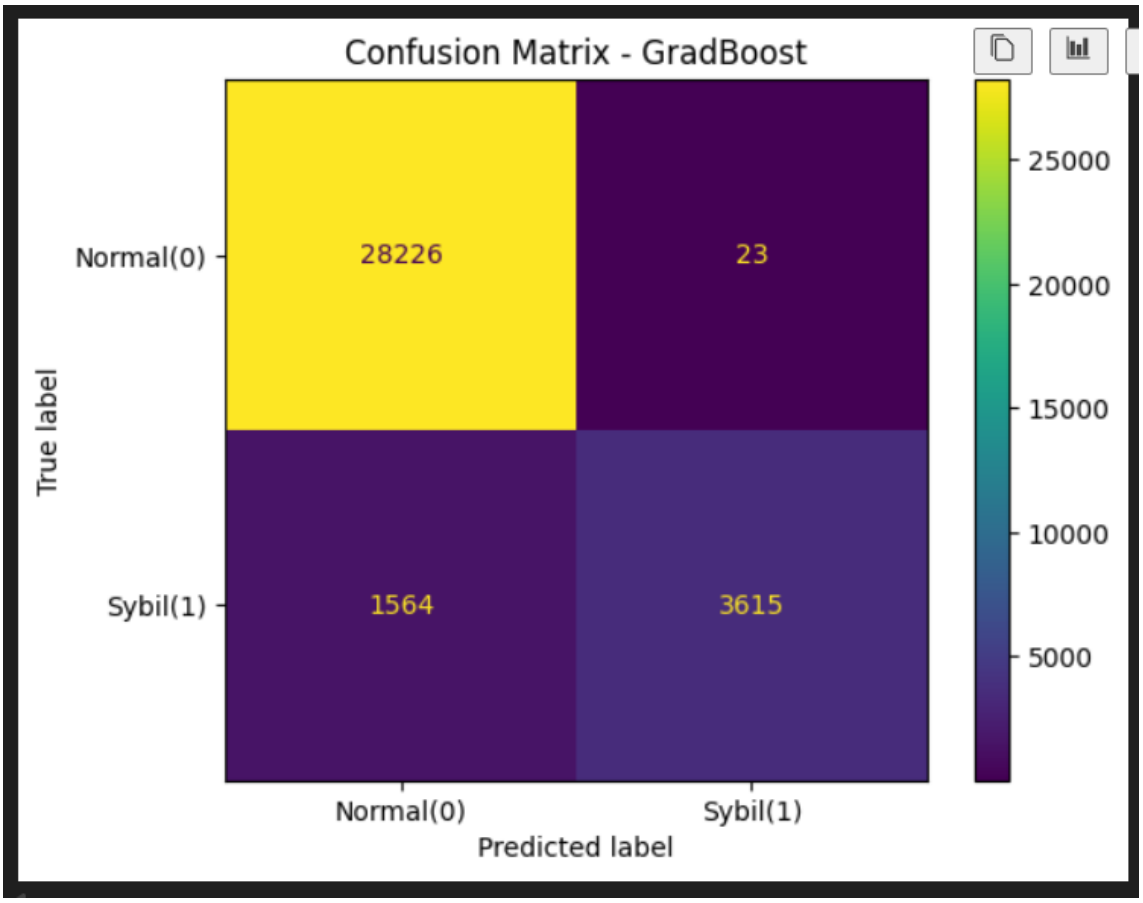



Confusion Matrix - RandomForest
Normal(0)
Sybil(1)
True label
24348
3901
152
5027
Normal(0)
Sybil(1)
Predicted label
20000
15000
10000
5000

GradBoost

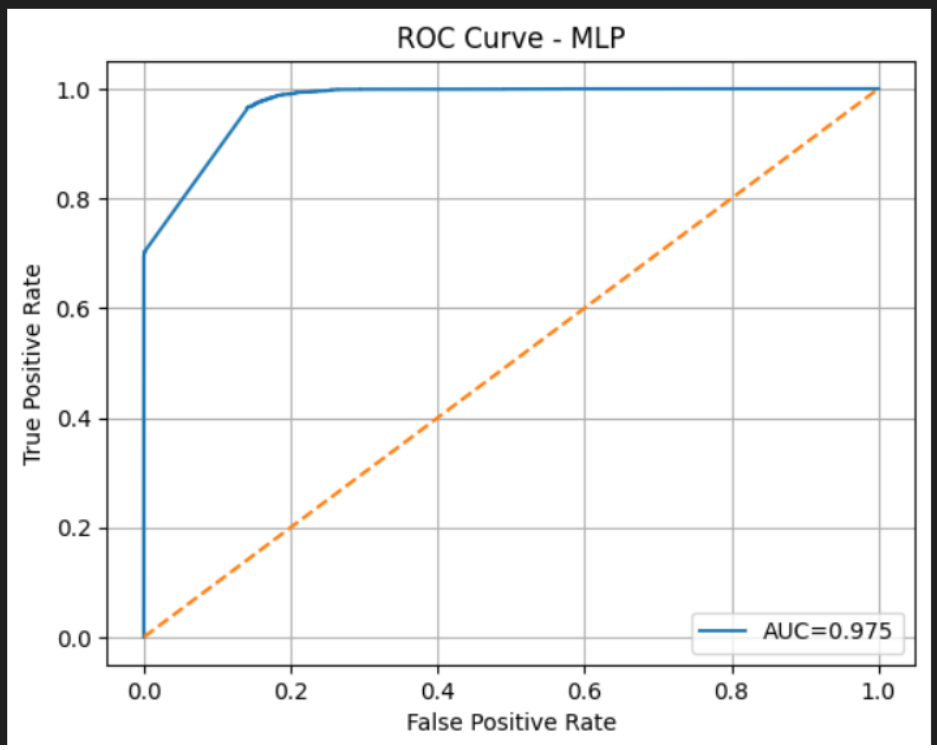
```
MODEL: GradBoost
              precision    recall  f1-score   support

           0     0.9475    0.9992    0.9727     28249
           1     0.9937    0.6980    0.8200      5179

    accuracy                         0.9525     33428
   macro avg     0.9706    0.8486    0.8963     33428
weighted avg     0.9547    0.9525    0.9490     33428
```

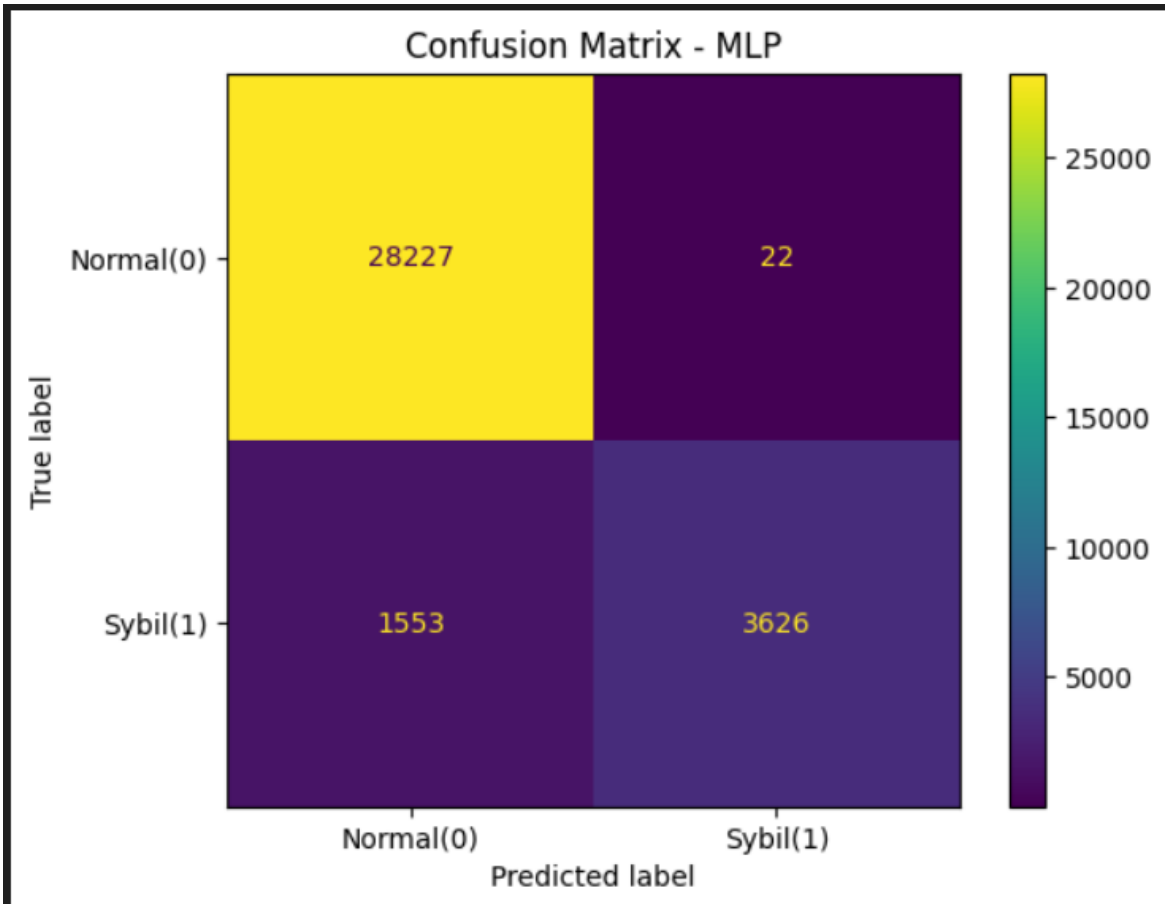



Confusion Matrix - GradBoost
Normal(0)
Sybil(1)
True label
28226
23
1564
3615
Normal(0)
Sybil(1)
Predicted label
25000
20000
15000
10000
5000

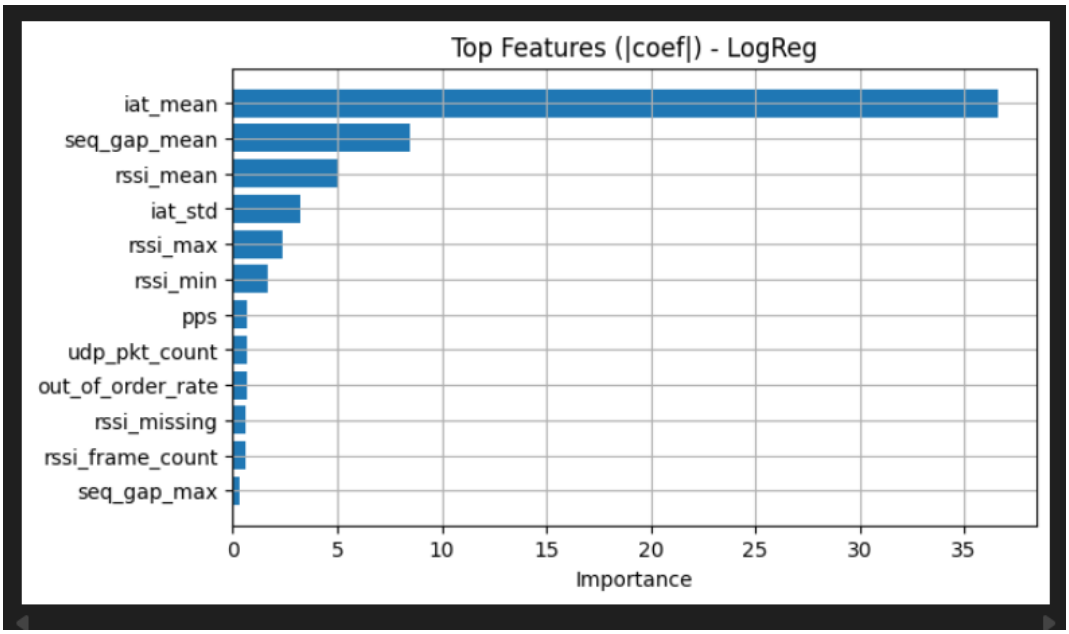
MODEL: MLP

```
MODEL: MLP
              precision    recall  f1-score   support

           0     0.9479    0.9992    0.9729     28249
           1     0.9940    0.7001    0.8216      5179

    accuracy                         0.9529     33428
   macro avg     0.9709    0.8497    0.8972     33428
weighted avg     0.9550    0.9529    0.9494     33428
```

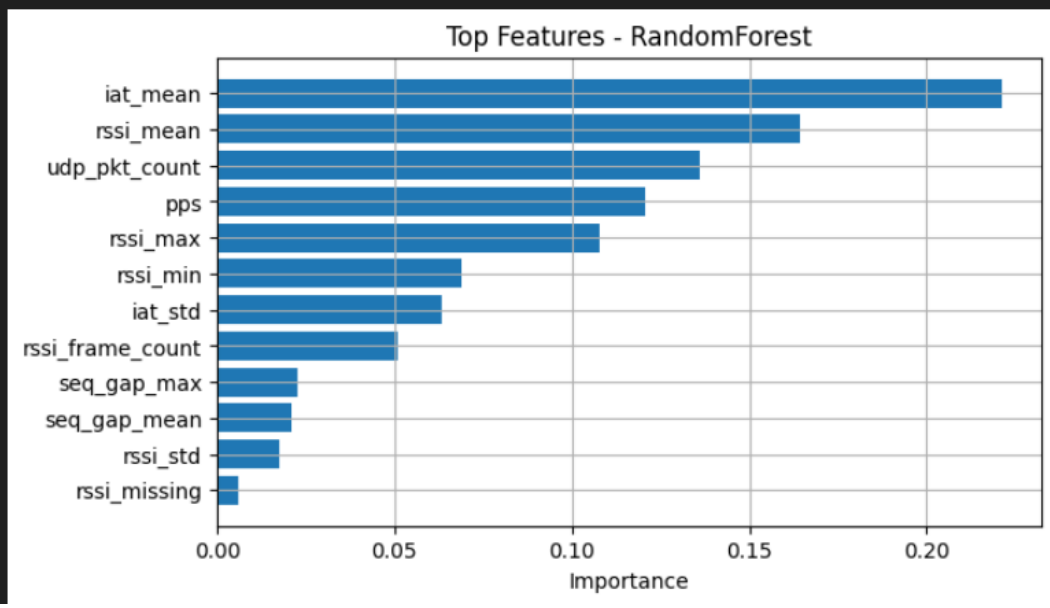




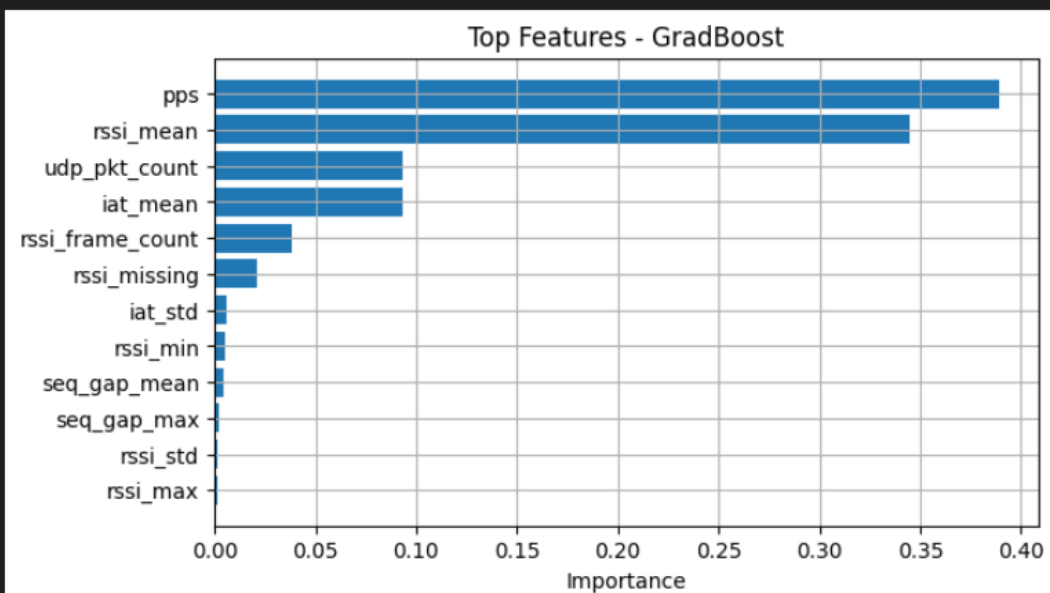
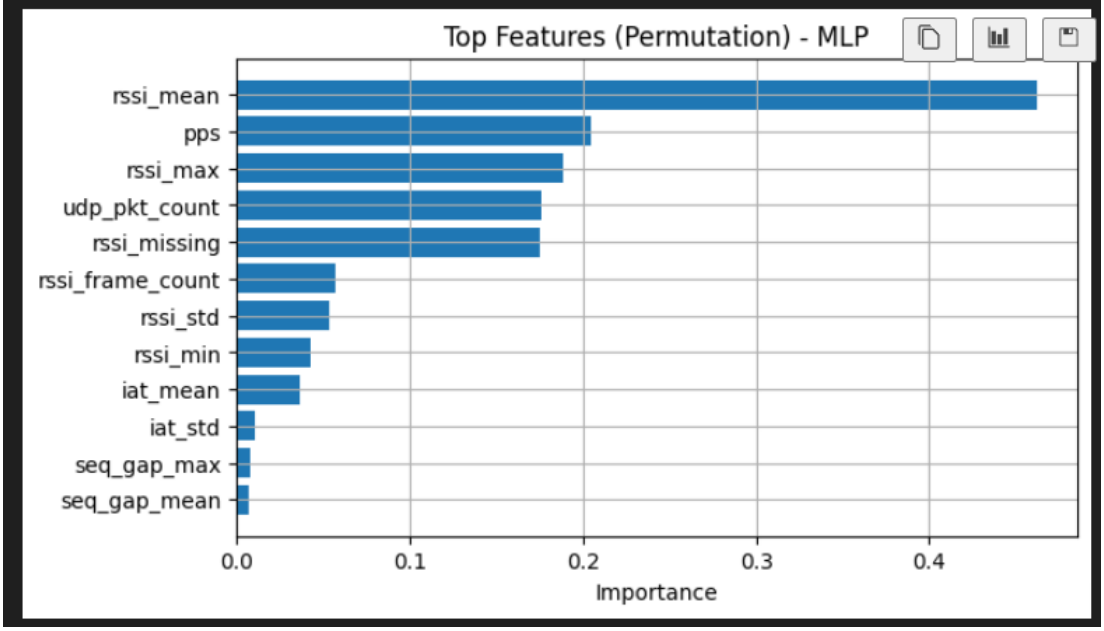
Feature importance



```
--- Feature importance for RandomForest ---
```




```
--- Feature importance for GradBoost ---
```



```
--- Feature importance for MLP ---
```